Travel-cost method

The travel-cost method (TCM) is used for calculating economic values of environmental goods. Unlike the contingent valuation method, TCM can only estimate use value of an environmental good or service. It is mainly applied for determining economic values of sites that are used for recreation, such as national parks. For example, TCM can estimate part of economic benefits of coral reefs, beaches or wetlands stemming from their use for recreational activities (diving and snorkelling/swimming and sunbathing/bird watching). It can also serve for evaluating how an increased entrance fee a nature park would affect the number of visitors and total park revenues from the fee. However, it cannot estimate benefits of providing habitat for endemic species.

TCM is based on the assumption that travel costs represent the price of access to a recreational site. Peoples' willingness to pay for visiting a site is thus estimated based on the number of trips that they make at different travel costs. This is called a revealed preference technique, because it 'reveals' willingness to pay based on consumption behaviour of visitors.

The information is collected by conducting a survey among the visitors of a site being valued. The survey should include questions on the number of visits made to the site over some period (usually during the last 12 months), distance travelled from visitor's home to the site, mode of travel (car, plane, bus, train, etc.), time spent travelling to the site, respondents' income, and other socio-economic characteristics (gender, age, degree of education, etc). The researcher uses the information on distance and mode of travel to calculate travel costs. Alternatively, visitors can be asked directly in a survey to state their travel costs, although this information tends to be somewhat less reliable. Time spent travelling is considered as part of the travel costs, because this time has an opportunity cost. It could have been used for doing other activities (e.g. working, spending time with friends or enjoying a hobby). The value of time is determined based on the income of each respondent. Time spent at the site is for the same reason also considered as part of travel costs. For example, if respondents visit three different sites in 10 days and spend only 1 day at the site being valued, then only fraction of their travel costs should be assigned to this site (e.g. 1/10). Depending on the fraction used, the final benefit estimates can differ considerably.

Two approaches of TCM are distinguished – individual and zonal. Individual TCM calculates travel costs separately for each individual and requires a more detailed survey of visitors. In zonal TCM, the area surrounding the site is divided into zones, which can be either concentric circles or administrative districts. In this case, the number of visits from each zone is counted. This information is sometimes available (e.g. from the site management), which makes data collection from the visitors simpler and less expensive.

The relationship between travel costs and number of trips (the higher the travel costs, the fewer trips visitors will take) shows us the demand function for the average visitor to the site, from which one can derive the average visitor's willingness to pay. This average value is then multiplied by the total relevant population in order to estimate the total economic value of a recreational resource.

TCM is based on the behaviour of people who actually use an environmental good and therefore cannot measure non-use values. This method is thus inappropriate for sites with unique characteristics which have a large non-use economic value component (because many people would be willing to pay for its preservation just to know that it exists, although they do not plan to visit the site in the future).

The travel-cost method might also be combined with contingent valuation to estimate an economic value of a change (either enhancement or deterioration) in environmental quality of the NP by asking the same tourists how many trips they would make in the case of a certain quality change. This information could help in estimating the effects that a particular policy causing an environmental quality change would have on the number of visitors and on the economic use value of the NP.

For further reading:

Ward, F.A., Beal, D. (2000) Valuing nature with travel cost models. A manual. Edward Elgar, Cheltenham.